

Jonathan Cruz

PRINCIPAL SOFTWARE ENGINEER · AI-AUGMENTED ENGINEERING

📍 SÃO PAULO, BRAZIL

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PROFILE

"Senior engineer with over 10 years building consumer and B2B platforms across web and mobile, with a strong focus on portfolio-level engineering excellence. Daily practitioner of AI-augmented engineering — sub-agent orchestration, spec-driven workflows, and agentic code generation/review pipelines. Deep TypeScript expertise across the stack with React/React Native, NestJS, Prisma, and Nx monorepos. Track record of mentoring engineers from junior to senior, modernizing legacy systems, and shipping reliable solutions in regulated and high-availability environments."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

FINTECH FOR VC FUND
ADMINISTRATION

Sydecar

FULL-STACK SOFTWARE
ENGINEER

AI- Augmented Engineering

AGENTIC WORKFLOWS

Daily user of agentic coding tools (Claude Code) with sub-agent orchestration, custom skills, project-level CLAUDE.md specifications, and pre/post-tool hooks. Apply spec-driven workflows for code generation, test scaffolding, code review, and architectural exploration — pairing AI execution layers with knowledge-centric project conventions to maintain quality at high iteration speed.

AI ENGINEERING AGENTIC WORKFLOWS SPEC-DRIVEN SUB-AGENTS

Nx Monorepo

CROSS-TEAM
ARCHITECTURE

Operate inside an Nx monorepo spanning dozens of NestJS libraries and React/Next.js apps. Drive cross-library architectural decisions (ADRs), enforce module boundaries, and coordinate changes across product teams to keep delivery flowing without blocking parallel workstreams.

NX MONOREPO ADR NESTJS

Complex Backend Systems

POSTGRES

Designed and implemented Enhanced Due Diligence with LP funding blocks, UBO verification, FinCEN integration, and CTA compliance. Built race-resistant flows with re-check inside to handle concurrent edits (409 EMAIL_NOW_EXISTS). Contributed to a proprietary double-entry ledger with append-only entries, deterministic idempotency keys, and event-driven posting.

POSTGRES PRISMA CONCURRENCY COMPLIANCE

Fault Tolerance

EVENT-DRIVEN

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup ().

BULLMQ REDIS OUTBOX EVENT-DRIVEN

External Integrations

CROSS-ORG
COLLABORATION

Built bidirectional QuickBooks invoice sync engine with state validation, auto-sync on approval, full E2E suite with Playwright and Cucumber BDD. Worked on Plaid Link integration with processor tokens powering ACH via Treasury Prime, including mock banking engine for local sandbox.

PLAID TREASURY PRIME QUICKBOOKS PLAYWRIGHT

Type-safe Contracts

DX EXCELLENCE

Designed (ADR) and implemented Billing V2 with quarterly fee scheduler and billing model versioning. Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas — improving developer experience and contract reliability end-to-end.

TS-REST ZOD TYPESCRIPT

MAY 2023 — PRESENT
SAN FRANCISCO, CA (REMOTE)

B2B OMNICHANNEL WHOLESALE Zax SR SOFTWARE ENGINEER JAN 2019 — MAY 2023 SÃO PAULO	Omnichannel Platform MOBILE + WEB Built from scratch a wholesale ordering app integrating mobile and web channels for business partners, with credit-limit validation and order-to-cash flows, using React Native, TypeScript, and GraphQL. REACT NATIVETYPESCRIPTGRAPHQL Scalable Cloud AWS Deployed full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless web apps using React and Next.js. Containerized services with Docker for consistent dev-to-prod parity. AWSNEXT.JSSERVERLESSDOCKER Engineering Excellence CI/CD Configured end-to-end delivery pipeline with GitHub Actions, Fastlane, and Code Climate for automated quality gates. Reduced release cycle from weeks to days. GITHUB ACTIONSFASTLANEDOCKER
CONSUMER MOBILE + WEB Netlolo SR SOFTWARE ENGINEER SEP 2018 — JAN 2019 SÃO PAULO	Consumer Product REACT NATIVE Built a mobile and web product bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum. Demonstrated interest in researching and shipping alternative/emerging web technologies. REACT NATIVETYPESCRIPTWEB3 Real-time Systems SDK Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend for low-latency real-time messaging across multiple client integrations. REACTREDUXXMPPJAVA Zero Downtime DOWNTIME Built CI/CD automations ensuring continuous deployment with no service interruption. PostgreSQL with Hasura as the GraphQL layer. POSTGRESQLHASURACI/CD
CONSUMER HEALTHCARE Escale SR SOFTWARE ENGINEER JAN 2018 — SEP 2018 SÃO PAULO	Mentorship ENGINEERING CULTURE Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture, coaching engineers across product teams. Established patterns for service boundaries and API contracts adopted across the engineering org. MENTORSHIPTECH TALKSMICROSERVICES Microservices Architecture MULTI-DB Built the product foundation with React, Redux, and a Node.js/TypeScript backend. Selected appropriate persistence per service responsibility — MySQL, PostgreSQL, and MongoDB — driving DB design and tuning decisions. REACTNODE.JSMONGODBCIRCLECI
FINTECH PraValer PL SOFTWARE ENGINEER JUN 2017 — JAN 2018 SÃO PAULO	Legacy Modernization NODE Migrated legacy PHP origination platform to a Node.js/Express backend for a student loans product with governmental funding, modernizing API surface while preserving downstream consumer compatibility. NODE.JSEXPRESSLEGACY MIGRATION Database PERFORMANCE TUNING Restructured PostgreSQL transactional tables with improved schemas and indexes, eliminating N+1 queries and reducing query times by an order of magnitude. MongoDB for request logs and unstructured data. POSTGRESQLMONGODBQUERY OPTIMIZATION

CORE SKILLS

AI ENGINEERING

Agentic workflows, Sub-agent orchestration, Spec-driven development, Prompt engineering, AI-augmented code review and testing

BACKEND

TypeScript, NestJS, Node.js, Prisma, PostgreSQL, ts-rest, Zod, GraphQL, Event-driven architecture

FRONTEND

React, Next.js, React Native, RxJS, TanStack Query, HTML, CSS preprocessors

RELIABILITY

BullMQ, Outbox pattern, Idempotency, Redis, Datadog

INFRASTRUCTURE

AWS, GCP, Docker, CI/CD, Nx Monorepo, Git

LEADERSHIP

Mentoring engineers (junior → senior), ADRs, Cross-team coordination, Tech Talks

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY · JAN 2016 — JUL 2018